

Qatar-1b

R-0603

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_231001 · created 2026-07-25T04:48:59+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460218.6871 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.021
Transit depth (Rp/Rs)^2	2.37 +/- 0.64 [%]
Orbital Inclination (inc)	84.86 +/- 1.8
Airmass coefficient 1 (a1)	1.126 +/- 0.021
Airmass coefficient 2 (a2)	-0.094 +/- 0.037
Scatter in the residuals of the lightcurve fit is	1.83 %
Best Comparison Star	#1 - [224, 381]
Optimal Aperture	3.38
Optimal Annulus	11.52
Transit Duration (day)	0.0715 +/- 0.0096

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.154 $\pm$ 0.021 vs 0.1463 (deviation 0.35 σ)
Duration fit vs published	0.0715 d vs 0.0692 d (deviation 0.23 σ)
Mid-time O-C	-6.6 min
Depth SNR	7.1
Residual scatter	1.83 %

Light curve

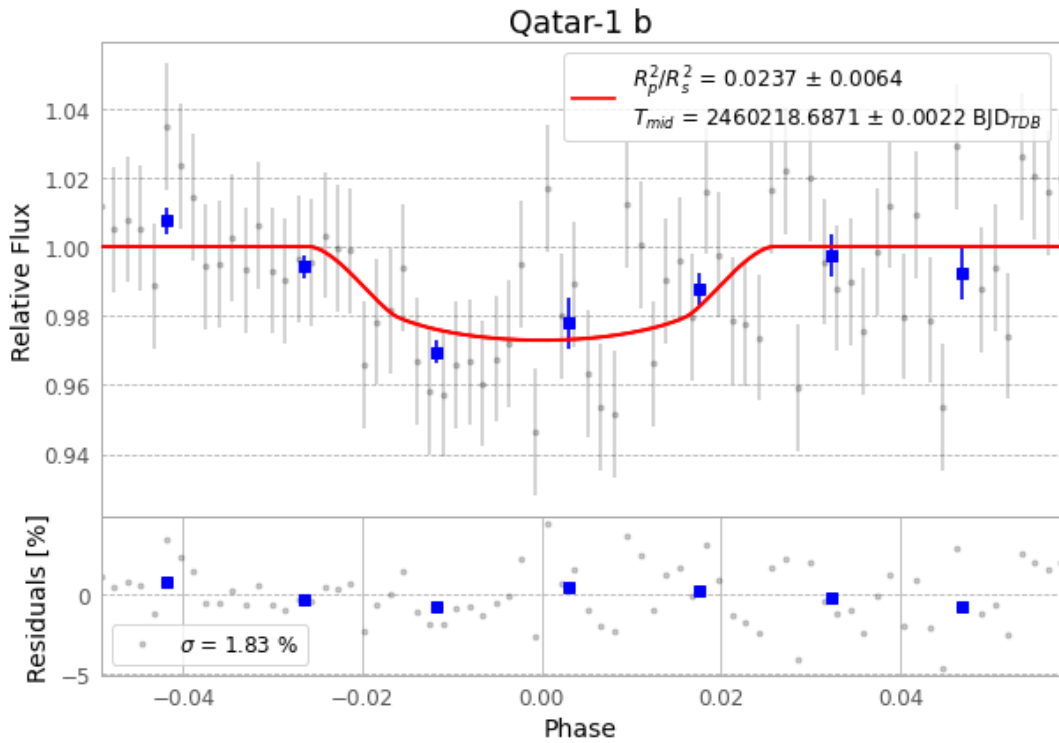


Figure 1 — FinalLightCurve_Qatar-1 b_2023-09-30.png (SHA-256 ccd5cf64ad29581b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [263, 209], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 7bf5e336dc171c4d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

74 FITS frames (49 MB), Qatar-1231001024516.FITS → Qatar-1231001062415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-231001.log (fa2995eb0c1a...), FinalLightCurve_Qatar-1 b_2023-09-30.png (ccd5cf64ad29...), FinalLightCurve_Qatar-1 b_2023-09-30.csv (18b012cf2456...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)